MILITARY SPECIFICATION

PRIMER, PERCUSSION, M81
PARTS FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the parts for one type of Initiation Element designated as Primer, Percussion, M81.

2. APPLICABLE DOCUMENTS

2.1 The following docuements of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition & Special Weapons, General Specification for

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-1168- Ammunition Lot Numbering
MIL-STD-1169- Packaging, Packing and Marking in Shipment of Inert Ammunition Components
MIL-STD-1235- Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

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DRAWINGS

ARMY

8838012 - Cup, Battery
8838015 - Plug, Firing
8838085 - Head
8838157 - Plug, Closing
8848623 - Body

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with applicable drawings and specifications

3.2 Parts.-The parts shall comply with all requirements specified on Drawings (dwgs) 8838012, 8838015, 8838085, 8838157 and 8838623, as applicable, and with all requirements specified in applicable specifications.

3.3 Hydrostatic Test.-The tubing to be fabricated into primer bodies shall withstand the application of a minimum (min.) hydrostatic pressure without rupture or indication of flaws, as evidenced by leaks and without permanent deformation beyond drawing dimensions when tested as specified in 4.4.1 (see 6.3).

3.4 Workmanship.-All parts shall be fabricated and finished in a thorough, workmanlike manner. They shall be clean and free of burrs, sharp edges, unblended radii, surface defects, chips, dirt, grease, rust and other foreign matter. The cleaning method used shall not be injurious to any parts nor shall the parts be contaminated by the cleaning agent. All required marking shall be neat and sharply defined.

3.5 First Article inspection.-This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contact.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection.-Unless otherwise specified in the contract or purchase order, the supplier is

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responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. In addition, the Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.2 First Article Inspection

4.2.1 Submission.-The contractor shall submit a first article sample designated by the contracting officer for evaluation in accordance with the provisions of 4.2.2. The first article sample shall consist of 25 each individual components, which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials including packaging and packing, shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirement of the contract and submit a record of this inspection with the sample, including statements of findings for materials, processes, and tests. A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer. A first article sample or portion thereof as directed by the contracting officer shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government.

4.2.2 Inspection to be performed.-Assemblies and components will be subjected by the Government to any or all of the examinations or tests specified in this specification and any or all requirements of the applicable drawings.

4.2.3 Rejection.-If any assembly or component fails to comply with any of all of the applicable requirements, the first article sample shall be rejected. The Government reserves the

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right to terminate its inspection upon any failure of assembly component in the sample to comply with any or all of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article sample or portion thereof. Until a first article sample is accepted, the contractor is, in no way, authorized by the Government to initiate regular production unless otherwise directed by the contracting officer.

4.3 Inspection Provisions

4.3.1 Inspection Lot Formation.-The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions, and complies with the provisions for submission of product as specified in MIL-STD-105. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168.

4.3.2 Examination.-Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL-STD-105 except that inspection for critical defects, when listed, shall be 100 percent. Continuous sampling plans in accordance with Standard MIL-STD-1235 may be used if approved by the procuring activity. Sampling plans shall be applied to the individual characteristics listed using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

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4.3.2.1 Cup, battery (see dwg. 8838012).

Categories	Defects	Method of Inspection
Critical:	None defined	
Major:		
101.	Outside diameter, min.	Gage CD
102.	Thickness through bottom, maximum (max.)	Gage CD
103.	Length	Gage CD
Minor:		
201.	Diameter of hole	Gage CD
202.	Thickness through bottom, min.	Gage CD
203.	Inside diameter	Gage CD
204.	Outside diameter, max.	Gage CD
205.	Flat missing from surface at open end	Visual
206.	Evidence of poor workmanship (see 3.4)	Visual

4.3.2.2 Plug, firing (see dwg. 8838015).

Categories	Defects	Method of Inspection
Critical:	None defined	
Major:		
101.	Large diameter, min.	Gage CD
102.	Length and profile of point	Gage CD
103.	Length of pilot, min.	Gage CD
104.	Total length	Gage CD
Minor:		
201.	Large diameter, max.	Gage CD
202.	Length of pilot, max.	Gage CD
203.	Diameter of pilot	Gage CD
204.	Finish improper	Visual
205.	Evidence of poor workmanship (see 3.4)	Visual

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4.3.2.3 Plug, closing (see dwg. 8838157).

Categories	Defects	Method of Inspection
Critical:	None defined	
Major:		
101.	Pitch diameter of thread, min.	Gage CD
102.	Major diameter of thread, min.	Gage CD
Minor:		
201.	Pitch diameter of thread, max.	Gage CD
202.	Major diameter of thread, max.	Gage CD
203.	Total length	Gage CD
204.	Width of wrench slot	Gage CD
205.	Evidence of poor workmanship (see 3.4)	Visual
206.	Protective coating damaged exposing base metal	Visual

4.3.2.4 Head (see dwg. 8838085).

Categories	Defects	Method of Inspection
Critical:	None defined	
Major:		
101.	Pitch diameter of body thread	Gage CD
102.	Major diameter of body thread, min.	Gage CD
103.	Pitch diameter of case thread	Gage CD
104.	Major diameter of case thread	Gage CD
105.	Diameter of body pilot, max.	Gage CD
106.	Diameter of battery cup cavity	Gage CD
107.	Diameter of percussion element cavity	Gage CD
108.	Depth of percussion element cavity	Gage CD
109.	Thickness of flange, max.	Gage CD
110.	Crack or split	Visual
111.	Flash hole missing or obstructed	Visual

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Minor:

201.	Major diameter of body thread, max.	Gage CD
202.	Diameter of flange	Gage CD
203.	Depth of battery cup cavity	Gage CD
204.	Diameter of body pilot, min.	Gage CD
205.	Concentricity between percussion element cavity and flash hole	Gage CD
206.	Concentricity of percussion element cavity with case thread	Gage CD
207.	Concentricity of flange with case thread	Gage CD
208.	Diameter of chamfer at mouth of percussion element cavity	Gage CD
209.	Diameter of flash hole	Gage CD
210.	Length to shoulder above body thread	Gage CD
211.	Length of body thread undercut	Gage CD
212.	Length of case thread undercut	Gage CD
213.	Length from lead end to case thread shoulder	Gage CD
214.	Diameter of shoulder between threads	Gage CD
215.	Distance between wrench slots	Gage CD
216.	Depth of wrench slots, min	Gage CD
217.	Total length	Gage CD
218.	Depth of flash cavity	Gage CD
219.	Diameter and profile of flash cavity	Gage CD
220.	Thread undercut missing	Visual
221.	Radius missing	Visual
222.	Finish improper	Visual
223.	Evidence of poor workmanship (see 3.4)	Visual

4.3.2.5 Body (see dwg. 8848623).

Categories	Defects	Method of Inspection
Critical:		
1.	Flash hole missing	Gage CD
2.	Crack or split	Visual

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Major:

101.	Pitch diameter of head thread	Gage CD
102.	Minor diameter of head thread, max.	Gage CD
103.	Location of first two pairs of holes	Gage CD
104.	Diameter of holes	Gage CD
105.	Pitch diameter of closing plug thread	Gage CD
106.	Minor diameter of closing plug thread, max.	Gage CD

Minor:

201.	Wall thickness, min.	Gage CD
202.	Length, min.	Gage CD
203.	Length of head thread, min.	Gage CD
204.	Inside diameter, min.	Gage CD
205.	Outside diameter, max.	Gage CD
206.	Depth of thread counterbore, max.	Gage CD
207.	Diameter of thread counterbore	Gage CD
208.	Length of closing plug thread, min.	Gage CD
209.	Protective coating damaged exposing base metal	Visual
210.	Evidence of poor workmanship (see 3.4)	Visual

4.3.3 Testing

4.3.3.1 Hydrostatic test of primer body (see 3.3) - Critical Defect - CD.-This test shall be performed as specified in 4.4.1, using test equipment specified in 4.3.4. Any tubing which fails to meet the specified requirement shall be classed defective and removed from the lot.

4.3.4 Inspection Equipment.-The inspection equipment required to perform the examinations and tests prescribed in this section is identified under the "Method of Inspection" headings in 4.3.2 and 4.3.3 (see 6.2).

4.4 Test methods and procedures

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4.4.1 Hydrostatic test.-Prior to machining, the tubing for primer fabrication shall be assembled into the test equipment specified in 4.3.4 and shall be tested for not less than five seconds at not less than the hydrostatic pressure calculated from the following formula:

$$P = \frac{2\ St\ X.9}{D}$$

where:

P = test pressure in pounds per square inch (p.s.i.)
S = min. specified yield strength of tubing in psi.
t = specified min. wall thickness in inches
D = specified max. diameter in inches

Observation shall be made for rupture, indications of flaws as evidence by leakage and for permanent deformation beyond drawing requirements.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking

5.1.1 Level C.-Packing and marking shall be in accordance with Specification MIL-STD-1169.

6. NOTES

6.1 Ordering Data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

c. First article inspection provisions.

6.2 Inspection Equipment Designs.-Inspection equipment designs are of two types - Government Special Inspection Equipment (SIE) designs and contractor designs (CD). SIE designs are designated by drawing numbers under the "Method of Inspection" heading Section 4, design responsibility for all other inspection equipment is assigned to the contractor. The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.2.2 and 6.2.3 should he elect to do so.

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6.2.1 SIE Designs.-SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

b. A source control drawing or a specification control drawing as defined in MIL-STD-100.

c. An envelope drawing, as defined in MIL-STD-100, meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.2.2 Contractor Designs.-Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public). Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment, or with SIE designs if not otherwise covered thereby (see 6.2.1c). Designs shall be of the category and form (per MIL-D-1000) specified in the Contract Data Requirements Lists (DD Form 1423). The contractor is referred to MIL-HDBK-204 "Inspection Equipment Design" for guidance. The item detail specification number, paragraph number , and defect number from Section 4 shall be referenced on each contractor design together with the component or assembly drawing number, revision letter and date to which the specific design applies.

6.2.3 Submission of Designs for Approval.-Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements Lists (DD Form 1423). Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs

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and the required delivery schedule as stipulated in the contract. Designs shall be submitted for approval to Commander, Picatinny Arsenal, ATTN: SARPA-QA-T-T, Dover, New Jersey 07801.

6.3 Hydrostatic Test.-If the primer tubing procured for the bodies manufactured in accordance with this specification has been previously subjected to the hydrostatic test a second hydrostatic test will not be required.

Custodian:
Army-PA

Preparing Activity:
Army-PA

Project Number: 1390-A088

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